

E-commerce Sales Performance and Profitability Analysis

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Abstract

This research paper provides a comprehensive study of e-commerce retail data to optimize sales and profit. Utilizing a dataset of over 9,000 transactions, the study analyzes performance across various categories. The primary findings indicate a total revenue of \$2,297,200.86 and total profit of \$286,397.02. The implementation of a centralized dashboard allows for better 'Ease of Use' in tracking business health.

I. Introduction

With the surge in online retail, organizations require robust systems to interpret transaction data. This paper explores the 'Superstore' dataset to understand consumer trends and product performance. The focus is on creating a user-friendly interface for stakeholders to make informed decisions.

II. Ease of Use

The proposed analysis system prioritizes ease of use by aggregating complex datasets into a single-view dashboard. By utilizing Pivot Tables and slicers in Excel, users can filter data by Year, Region, and Category without technical expertise. This simplifies the decision-making process for department managers.

III. Figures and Tables

Table I below summarizes the performance of the core product categories. The data shows Technology contributes significantly to profit margins.

Category	Total Sales (\$)	Total Profit (\$)
Furniture	741,999.80	18,451.27
Office Supplies	719,047.03	122,490.80
Technology	836,154.03	145,454.95

Fig 1. Sales Distribution Analysis (Placeholder for Dashboard Image)

Figures in this report represent the dynamic nature of the dashboard, including map charts identifying sales density and waterfall charts highlighting category-wise profit contribution.

IV. Project Objectives

- Quantify KPIs: Sales (\$2,297,200.86), Profit (\$286,397.02), and Quantity (37,873).
- Identify high-performing categories to streamline supply chain efforts.
- Map geographical distribution to locate top states like California.
- Analyze temporal trends (YoY) from 2011 to 2014.
- Optimize inventory by tracking top-selling sub-categories like Binders and Chairs.

V. Conclusion

In conclusion, the analysis successfully transforms raw data into a strategic asset. The use of tables and figures enhances the interpretability of results, ensuring that profitability challenges in segments like Furniture are addressed through data-driven action.

References

[1] IEEE Conference Format Guidelines, 2023.

[2] Superstore Sales Dataset (2011-2014).

[3] LinkedIn : https://www.linkedin.com/posts/yuvrajpratapsingh2005_dataanalytics-ecommerce-datavisualization-7450172620945391616

[4] Dataset : <https://docs.google.com/spreadsheets/d/1L6aBX0uNlzKiJb7JHdkNUile1s9CI4r/edit?gid=1589100670>